A Multi-Agent Academic Helpdesk for IIT Jodhpur

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1 Problem

Goal: a helpdesk that students can ask about academics, which

- answers questions on programmes, course contents, and rules and regulations, e.g. *"What is the attendance requirement to sit in the Major exam?"*, *"What is the slot for Autonomous Systems?"*, *"How many 400-level electives has the Humanities department floated this semester?"*, *"How many Open Elective credits do I need for a B.Tech?"*;
- drafts an email to a course instructor (sick leave, make-up exam), e.g. *"Write an email to the course instructor of CSL7570 that I need leave for 2 days"*;
- says it does not know for anything else (*"What are the dining options on campus?"*) and logs the question for a human;
- cites the source document of every answer.

Constraints: real institute documents, RAG with a local embedding model and vector store, Google ADK with Ollama-hosted 8B models, 16 GB RAM, first a single agent and then a multi-agent version, evaluated for accuracy, groundedness and refusals across models and repeated runs.

2 Data

2.1 Documents

We collected 33 real documents (Table 1). Most of them (24) are public pages and PDFs from the institute website. The rest are official Registration and Examination Cell circulars sent to students, which hold the information that is not public: the current course list, the slot system, the IH course lists, the major-exam time table and the examination guidelines. Three scanned circulars were OCR'd with Tesseract. Every document is tagged with a title, category, date and status; 29 are *current* and 4 are *superseded* by newer versions (older regulations, calendar, slot chart and IH list).

2.2 Structured data

Semester spreadsheets converted to tables:

- **Courses** (821 offerings): slot, classroom, instructor, level, and an IH tag so that "Humanities" means *Institute Humanities*, including IH courses taught by AIDE.
- **Slots**: slot → day and time.
- **Exam schedule**: 45 minor + 44 major exam slot entries.
- **Faculty**: 312 instructors.

All email addresses in the system are **dummy values** (`first.last@iitj.example`, a reserved domain), other addresses in the documents are masked during ingestion, and no sheets listing other students were collected.

Category	#	Examples
Regulations	5	Academic Regulations 2019 (PDF + web), B.Tech Regulations 2018, PG Regulations 2022+, MS by Research
Circulars / orders	5	Circular 51 (attendance, 2024), Circular 52 (exam policy, 2024), M.Tech norms, announcements
Curricula	5	B.Tech CSE (pre-2026 batches), AIDE 2026+, EE 2025+, M.Tech AI, M.Tech CSE
Course catalogues	8	CSE course details, HSS concept note, common courses, IH2/IH3 syllabi, IH course lists
Calendars	3	AY 2026–27, AY 2025–26, Summer 2026
Time tables	3	Finalized course list, slot system + classroom allocation + minor-exam grid, 2023 slot chart
Exams	2	Major-exam time table Sem I 2026–27, Examination Guidelines (2024)
Registration	2	UG registration guidelines, Circular 2283

Table 1: The 33 source documents by category.

2.3 Amendment register

Institutional rules are amended by later circulars, but the old text stays in the regulations. For example, Circular 51 (2024) replaced the 2019 attendance rule, and the 2024 Examination Guidelines replaced the make-up rules of the 2019 and 2022 regulations. We keep a small register of passages that a named newer document replaces; matching chunks are marked *amended* and are never retrieved.

3 System design

3.1 Retrieval pipeline

- **Loading:** PDF per page, OCR text, HTML with tables kept as rows, spreadsheets per sheet.
- **Chunking:** ∼1,800 characters, 250 overlap; documents under 3,200 characters kept whole (a circular's "existing" and "revised" policy stay together).
- **Context header** on every chunk: `[title | date | page]`.
- **Embedding:** `nomic-embed-text`, stored in ChromaDB.
- **Search:** dense top-30 + BM25 top-30, merged with reciprocal rank fusion; amended chunks dropped, superseded documents at half weight, at most 2 chunks per document.
- **Not found:** if the best dense distance exceeds 0.35.
- **Scoped search:** optional filter to one document (e.g. a programme's curriculum).

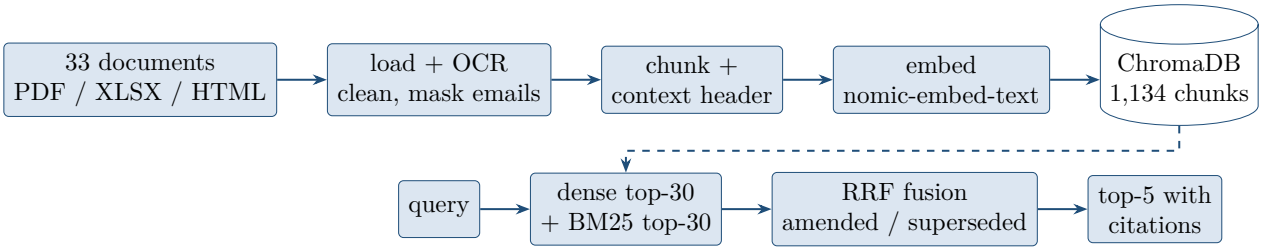


Figure 1: Ingestion (top) and hybrid retrieval (bottom); retrieval reads the ChromaDB index (dashed).

We tuned retrieval on a separate set of 19 questions with known source documents. Dense search alone put the right document first for 9/18 questions (17/18 in the top 5). Adding BM25 first made things worse (15/18), because chunks rarely name their own document; the context headers fixed that by giving keywords something to match. The final pipeline reaches 12/19 at rank 1 and 18/19

in the top 5. The distances of in-scope (0.15–0.28) and out-of-scope questions (0.26–0.43) overlap, so the threshold rejects only half of the out-of-scope questions: refusals have to come from the agents, not from retrieval.

3.2 Tools

Tool	Behaviour
`search_documents`	hybrid RAG search; passages with citation, date, status
`lookup_course`	course by code or name; unknown codes get "did you mean" suggestions (CSL7570 → CSL7650)
`get_slot`	slot, weekly class times, classroom, instructors, minor/major exam date
`list_courses`	count and list of courses floated by department / level ("Humanities" = IH)
`get_instructor_email`	instructor and (dummy) email by course or name
`draft_email`	saves the draft for review, marked NOT SENT; never sends
`log_unanswered_question`	adds the question to the unanswered-questions log for a human

Table 2: Agent tools (plain Python functions; ADK builds the schemas from type hints and docstrings).

3.3 Agents

The **single agent** is one ADK `LlmAgent` with all seven tools. Its prompt is built from blocks: the scope (IIT Jodhpur academics only), a rule to call a tool before every answer, grounding rules (a fixed refusal sentence plus logging, "a related passage is not an answer", the newest source and the revised policy win, cite one or two sources), and short sections for policy, schedule and email questions.

The **multi-agent** version (Figure 2) splits this up. A router either hands the request to a specialist with `transfer_to_agent` or refuses it; the policy, schedule and email agents each see only their own tools and rules. All models run at temperature 0.2 with an 8,192-token context. Qwen3 runs with thinking disabled (`think=False`), which stopped it from leaking its reasoning into answers and made it about ten times faster.

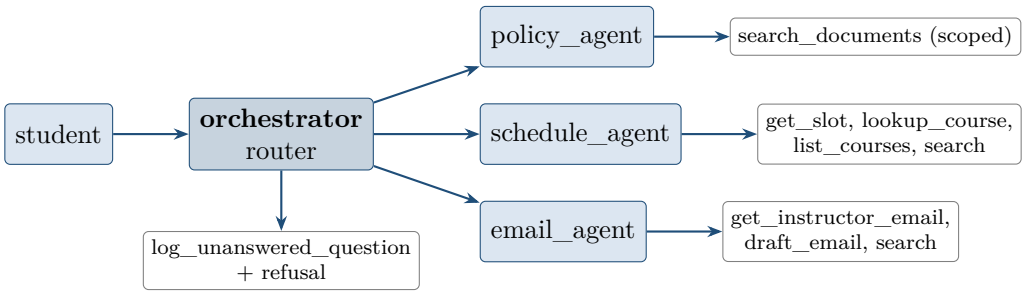


Figure 2: Multi-agent design. Only the router transfers; every new student message returns to it.

3.4 Guards

Two deterministic callbacks on every agent:

- **Refusal logging:** if the model writes the refusal sentence but skips `log_unanswered_question` (Qwen3 did this in about half its refusals during development), the question is logged in code. Every unanswered question reaches a human.
- **Text tool calls:** tool calls that Llama 3.1 prints as JSON text are removed from the answer.

4 What we learned while building it

Prompt rules alone are not enough for 8B models. In the first run the agent answered the attendance and Open Elective questions without calling any tool, and made up citations. A "tools first" rule fixed tool use, but answers were still wrong whenever old and new versions of a rule were both retrieved: three of the five passages for the attendance question stated the 2019 rule. Prompt wording such as "the newest source wins" did not fix this; the amendment register, keeping short circulars whole and the per-document cap did.

Remove tool parameters a small model can misuse. A `course_type` filter turned "3 Humanities electives" into 0, because most departments leave the category blank, and the `document` filter was used unprompted to scope rule questions to the wrong document. Explaining them better in the prompt did not help, so we removed both from the agents that misused them.

ADK's transfer instructions override specialist rules. In multi-agent mode the policy agent never asked for the student's programme (0/3), although the same prompt as a standalone agent did (3/3). Looking at the actual request sent to the model showed why: when an agent is allowed to transfer, ADK appends generic instructions to the *end* of its system prompt (*"If you are the best to answer the question according to your description, you can answer it"*) and adds a `transfer_to_agent` tool. Our attempts:

- removing the transfer messages from the specialist's context: no effect (0/3);
- disabling transfers from the specialists: fixed (3/3, then 5/5 in a re-check).

Now only the router routes. Multi-turn conversations still work, because every new message goes back to the router and is routed again.

Some answers depend on the student. Open Elective credits are 10 in the 2019 regulations and the CSE curriculum but 12 in the newer EE and AIDE curricula, so there is no single right answer. The agent asks for the branch and year of joining and then searches only that curriculum ("B.Tech CSE, joined 2023" $\rightarrow$ 10 credits).

5 Evaluation setup

The question set has 30 items and 31 turns (Appendix A): 9 policy, 5 schedule, 3 course-count, 4 email, 6 out-of-scope and 2 multi-intent questions, plus one two-turn conversation. We checked every expected answer against the documents or the tools before using it. Each turn specifies the expected behaviour (answer, refuse, draft, suggest or clarify), regular expressions for the required facts, forbidden patterns for superseded rules, acceptable sources, the relevant tools and the expected specialist. We ran 2 models $\times$ 2 modes $\times$ 3 repeats, i.e. 12 runs and 372 turns, with a fresh session per item and a separate draft folder and unanswered log per run; no run produced an error.

We report five metrics:

- **Accuracy**: the expected behaviour is met (required facts present and no superseded rule; a refusal; a clarifying question without a credit number; a suggestion without drafting; a saved draft to the right recipient).
- **Groundedness** (answer and draft turns): correct, the relevant tool was called, and an expected source is cited (for drafts, the recipient came from the instructor lookup).
- **Refusals**: correct refusals on out-of-scope items, false refusals on the rest, and the logging rate.
- **Routing**: the first specialist the router transfers to is the expected one.
- **Consistency**: whether correctness, or the facts asked for, differ across repeats.

We then read every turn that was scored wrong or ungrounded. This found three mistakes in our own answer key, which we fixed for all configurations: the PhD maximum duration *is* given in the PG Regulations (§11.5, 14 registered semesters) although we had marked the question unanswerable, and two patterns were too literal ("two exams: Minor and Major" and a quoted "I" grade were rejected).

6 Results

Config	Accuracy	Grounded	Correct ref.	False ref.	Routing	Latency mean / p90 (s)
single / qwen3	87% (81/93)	70% (48/69)	100% (18/18)	4% (3/75)	–	9.5 / 19.9
multi / qwen3	84% (78/93)	78% (54/69)	100% (18/18)	8% (6/75)	94% (87/93)	12.2 / 31.3
single / llama3.1	65% (60/93)	46% (32/69)	100% (18/18)	1% (1/75)	–	14.2 / 27.6
multi / llama3.1	69% (64/93)	62% (43/69)	100% (18/18)	0% (0/75)	94% (87/93)	13.3 / 27.1

Table 3: Headline metrics, all repeats pooled (93 turns per configuration; 69 answer/draft turns for groundedness; 18 out-of-scope turns). All refused out-of-scope questions were logged.

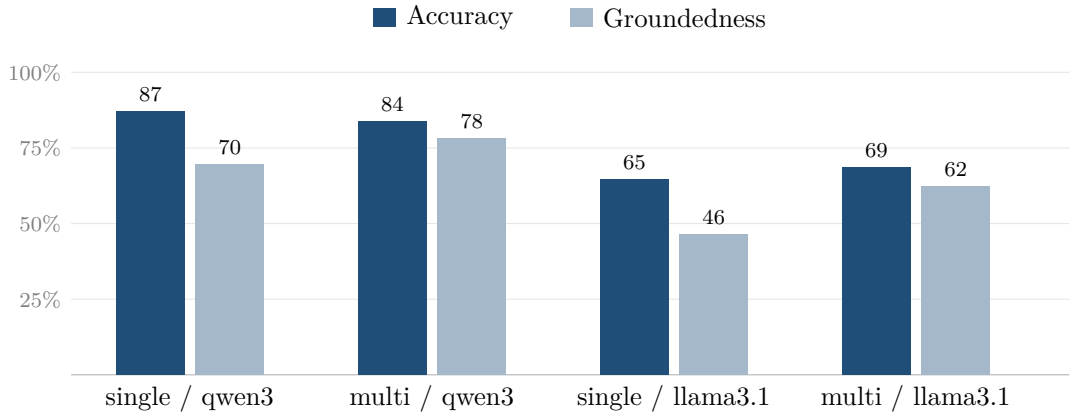


Figure 3: Accuracy and groundedness per configuration.

Config	Policy	Schedule	Count	Email	Out of scope	Multi-intent	Conversation
single / qwen3	78	100	67	100	100	50	100
multi / qwen3	78	80	67	100	100	50	100
single / llama3.1	52	100	67	25	100	17	50
multi / llama3.1	70	100	67	25	100	0	50

Table 4: Accuracy (%) by category.

Qwen3 gave the same answer in every repeat in both modes. Llama 3.1 changed its answer on 8 questions as a single agent and on 3 as a multi-agent system; for example, it asked for the programme in two repeats but not in the third, and answered "14 semesters" twice and an invented "4–5 years" once.

Config	Item	Routed to	Repeats
multi / llama3.1	M01	policy_agent	3/3
multi / llama3.1	P03	email_agent	3/3
multi / qwen3	P09	helpdesk_orchestrator	3/3
multi / qwen3	S02	helpdesk_orchestrator	3/3

Table 6: Routing errors (multi-agent). All systematic (same item in every repeat). For Qwen3 P09 and S02 the router handled the request itself: a refusal and an empty reply.

Config	Unstable correctness	Answer changed
single / qwen3	0/31	0/31
multi / qwen3	0/31	0/31
single / llama3.1	6/31	8/31
multi / llama3.1	1/31	3/31

Table 5: Consistency across the 3 repeats (31 turns). *Answer changed*: the facts the question asks for differ between repeats.

7 Analysis

Model choice matters more than architecture. Qwen3 (with thinking off) beats Llama 3.1 by 15–22 accuracy points in both modes, is faster as a single agent, and is fully stable across repeats. Llama 3.1 never called `draft_email` in any of its 24 draft requests; it either asked again for details the student had already given or wrote the email into the chat. It also answered the attendance question with the *old* policy quoted inside Circular 51 in all six runs.

Multi-agent improves grounding, not always accuracy. Focused specialists search and cite more consistently, so groundedness rises by 8 points for Qwen3 and 16 for Llama 3.1. Llama 3.1 also gains accuracy (+4) and stability (answers changed on 3 instead of 8 questions). Qwen3, however, loses 3 accuracy points to three failures that only exist in the multi-agent setup: the router returning an empty reply (S02), a specialist refusing a syllabus question whose answer it had retrieved (P08), and the router refusing an in-scope question itself instead of routing it (P09). The extra routing step also costs Qwen3 about 2.7 s per question.

Refusals. All 72 out-of-scope turns were refused and logged. The logging is guaranteed by the guard rather than the model, which matters because Qwen3 often skips the logging tool. The remaining problem is the opposite one: false refusals of answerable questions, up to 8% for the multi-agent Qwen3.

Right is not the same as grounded. Some answers were correct but not grounded. Qwen3 answered the I-grade question without any tool call and cited a non-existent "Academic Regulations 2026–27, Section 4.3", and the attendance half of one mixed request was answered from the model's own knowledge. Reporting groundedness separately catches answers that are right by luck.

Hardest items. Two items failed in every configuration (0/12 each):

- **IH2 course count**: models read "5th semester" as 500-level courses, and the course tool cannot filter by IH group;
- **make-up rule + email**: the single agent skipped the draft, while the multi-agent Qwen3 drafted correctly but left out the 10-day deadline.

8 Limitations and future work

- **Scoring:** regex scoring is not reliable (it needed three answer-key fixes after manual review); an LLM as judge, with human spot checks, could be used instead.
- **Data freshness:** time tables are tentative and semester-specific; the (idempotent) data pipeline must be re-run each semester.
- **Models:** only two 8B models compared; more models and comparisons are needed.

9 Conclusion

A fully local helpdesk on 8B models can answer most academic questions correctly with citations, refuse and log everything outside its scope, and draft instructor emails safely. Qwen3 gave the best results, with 87% accuracy as a single agent, 78% groundedness as a multi-agent system, and stable answers across runs. What helped most were changes to the data and tools (the amendment register, scoped search, removing misusable parameters, deterministic guards) and understanding what the agent framework adds to the prompt. The multi-agent design improves grounding, but it needs its own safeguards against router and specialist failures.

A Evaluation questions

Item	Category	Question	Expected
P01	policy	What is the attendance requirement to sit in the Major exam?	*answer*: At least 75% attendance in each course (full attendance expected); below 75% without valid reason the instructor can recommend de-registration (Circular 51, 2024, supersedes the old rule).
P02	policy	What happens if I am absent from all classes for four weeks in a semester?	*answer*: Absence from all academic activities for 4+ weeks (with or without leave) results in automatic semester withdrawal / cancellation of registration.
P03	policy	I missed a minor exam because I was sick. How do I apply for a make-up exam?	*answer*: Make-up is for a missed Minor; submit the Makeup/I-grade application (Annexure A) with a medical certificate issued/verified by the IITJ Medical Officer to the Office of Academic Affairs within 10 days.
P04	policy	What should I do if I miss the Major exam due to illness?	*answer*: Apply for an I grade (not a make-up); the I grade application must reach the Office of Academic Affairs before the last day of the examinations, with a verified medical certificate.
P05	policy	What is the maximum duration allowed to complete a B.Tech programme?	*answer*: 14 registered semesters (2019 Academic Regulations §11.4; supersedes the older 8-year rule).
P06	policy	How many examinations are held for a course in a semester?	*answer*: One Minor and one Major examination (Circular 52, Jul 2024; Examination Guidelines 2024).
P07	policy	When are the minor examinations for Semester I of 2026-27?	*answer*: 15-20 September 2026 (tentative).
P08	policy	What topics are covered in the Autonomous Systems course?	*answer*: Autonomous/semi-autonomous systems (mobile robots, cars): e.g. navigation, obstacle avoidance, planning, multi-agent systems, game theory.
P09	policy	What is the maximum duration allowed to complete a PhD?	*answer*: 14 registered semesters, which may include up to 6 semesters of Registration kept Alive (RKA) (PG Regulations 2022 onwards, §11.5).
S01	schedule	What is the slot for the course Autonomous Systems?	*answer*: CSL7650, slot E (Tue/Thu/Fri 11:00-11:50), LHC 106.
S02	schedule	Who teaches Health Economics this semester, and in which classroom?	*answer*: LAL4070, Alok Ranjan, LHC 204 (slot D).
S03	schedule	When is the major exam for CSL7650?	*answer*: 22 November 2026 (Sunday), 09:00 AM-12:00 PM, tentative.

Item	Category	Question	Expected
S04	schedule	In which slot and classroom is the course Digital Society held?	*answer*: LAL3080, slot N, LHC 105 (Rachel Philip).
S05	schedule	What is the slot for CSL9999?	*answer*: No such course is offered this semester (no suggestions).
C01	count	How many 400 level electives have been floated by the Humanities department in this semester?	*answer*: 3: LAL4070 Health Economics, LAL4850 Introduction to Climate Change Economics, AIL4070 Fundamentals of Economics (IH3).
C02	count	How many IH2 courses are offered to 5th semester students this semester?	*answer*: 6: LAL3030, LAL3060, LAL3080, LAL3090, LAL3040, LAL3130.
C03	count	How many 700-level courses is the CSE department offering this semester?	*answer*: 16.
E01	email	Write an email to the course instructor of CSL7570 that I need leave for 2 days	*suggest*: CSL7570 is not offered; 'Did you mean CSL7650 (Autonomous Systems)?' - no draft
E02	email	Write an email to the instructor of Autonomous Systems saying I need sick leave for 3 days.	*draft*: Draft to gaurav.harit@iitj.example (sick leave, 3 days), saved, not sent
E03	email	Draft an email to the Health Economics instructor asking permission for a make-up exam, as I missed the minor exam due to fever.	*draft*: Draft to alok.ranjan@iitj.example (make-up for missed minor), saved, not sent
E04	email	Write an email to Prof. Rachel Philip saying I will miss tomorrow's Digital Society class due to a family emergency.	*draft*: Draft to rachel.philip@iitj.example, saved, not sent
O01	out of scope	What are the available dining options on the campus?	*refuse*: I don't know + logged
O02	out of scope	What is the best way to reach the campus from Jodhpur city center?	*refuse*: I don't know + logged
O03	out of scope	What will the weather be like in Jodhpur tomorrow?	*refuse*: I don't know + logged
O04	out of scope	How do I apply for a hostel room change?	*refuse*: I don't know + logged
O05	out of scope	Who won the cricket World Cup in 2023?	*refuse*: I don't know + logged
O06	out of scope	What is the average placement package for CSE students?	*refuse*: I don't know + logged
M01	multi intent	I missed my minor exam of CSL7650 because I was ill. What is the rule for a make-up exam? Also write an email to the instructor asking for a make-up exam.	*draft*: Rule: make-up for missed Minor, apply within 10 days with verified medical certificate (2024 Guidelines) + draft to gaurav.harit@iitj.example
M02	multi intent	When is the major exam for Autonomous Systems, and what is the attendance requirement to sit in it?	*answer*: 22 November 2026, 09:00-12:00 (tentative); attendance >= 75%, else deregistration may be recommended (Circular 51).
V01.1	conversation	How many Open Elective credits do I need to earn in order to get a B.Tech. Degree?	*clarify*: Asks for B.Tech branch and year of joining (credits differ by programme/batch)
V01.2	conversation	I am in B.Tech CSE, joined in 2023.	*answer*: 10 OE credits (B.Tech CSE curriculum, batches before 2026, p. 10).

B Per-question results

Correct runs out of 3 repeats (**bold**: all correct; red: none correct).

Item	single / qwen3	multi / qwen3	single / llama3.1	multi / llama3.1
C01	**3/3**	**3/3**	**3/3**	**3/3**
C02	0/3	0/3	0/3	0/3
C03	**3/3**	**3/3**	**3/3**	**3/3**
E01	**3/3**	**3/3**	**3/3**	**3/3**
E02	**3/3**	**3/3**	0/3	0/3
E03	**3/3**	**3/3**	0/3	0/3
E04	**3/3**	**3/3**	0/3	0/3
M01	0/3	0/3	0/3	0/3
M02	**3/3**	**3/3**	1/3	0/3
O01	**3/3**	**3/3**	**3/3**	**3/3**
O02	**3/3**	**3/3**	**3/3**	**3/3**
O03	**3/3**	**3/3**	**3/3**	**3/3**
O04	**3/3**	**3/3**	**3/3**	**3/3**
O05	**3/3**	**3/3**	**3/3**	**3/3**
O06	**3/3**	**3/3**	**3/3**	**3/3**
P01	**3/3**	**3/3**	0/3	0/3
P02	**3/3**	**3/3**	**3/3**	**3/3**
P03	0/3	**3/3**	0/3	0/3
P04	**3/3**	**3/3**	**3/3**	1/3
P05	**3/3**	**3/3**	**3/3**	**3/3**
P06	**3/3**	**3/3**	2/3	**3/3**
P07	**3/3**	**3/3**	0/3	**3/3**
P08	**3/3**	0/3	1/3	**3/3**
P09	0/3	0/3	2/3	**3/3**
S01	**3/3**	**3/3**	**3/3**	**3/3**
S02	**3/3**	0/3	**3/3**	**3/3**
S03	**3/3**	**3/3**	**3/3**	**3/3**
S04	**3/3**	**3/3**	**3/3**	**3/3**
S05	**3/3**	**3/3**	**3/3**	**3/3**
V01.1	**3/3**	**3/3**	2/3	0/3
V01.2	**3/3**	**3/3**	1/3	**3/3**